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Economic evaluation of a roll-off trucking system removing forest biomass resulting from shaded fuelbreak treatments

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ABSTRACT

Shaded fuelbreak treatments involve removal of understory brush and small-diameter trees to reduce fire hazards by disconnecting the continuity of fuels. As a result of these treatments, woody biomass (referred to as slash) is piled throughout the treated stand and later burned. Mechanical removal of slash has not been successfully implemented in many areas due to limited accessibility to sites and the high costs associated with collection and transportation of slash. To address these issues, a roll-off truck paired with a small skid-steer loader was used to collect and transport slash to a centralized processing site where slash was ground as hog fuel for energy production. "Roll-off truck" refers to a straight frame truck configuration in which a 30.6-m^3 container is rolled onto and off the straight frame truck by means of a truck-mounted winch system. This study was designed to quantify the operational performance and costs of removing slash piles using a roll-off trucking system in mountainous conditions in northern California. The overall cost to collect and haul hand-piled slash was $26.81/tonne with 22% average moisture content or $34.37/bone dry metric ton. The roll-off trucking system should be used primarily for short hauling distances since trucking costs significantly increase with small increases in hauling distance due to slow traveling speeds and low slash weight being hauled. Financial analysis indicated that contractors can receive high rates of return on their invested capital after accounting for inflation and income taxes, but limited work opportunities are a concern for them.

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1. Introduction

A fuelbreak is a strategically located wide block, or strip, on which a cover of dense, heavy, or flammable vegetation has been changed to one of lower fuel volume or reduced flammability [4]. Shaded fuelbreaks are often created by altering surface fuels, increasing the height to the base of live crowns, and opening the canopy by removing trees [1]. This practice has been commonly implemented on National Forests in northern California to break up the continuity of fuels, improve firefighter safety, and confine wildfires to one watershed area. In mixed conifer vegetation types, a high level of crown closure (typically around 60%) is recommended in order to reduce re-growth of brush and small trees after the treatments. Chainsaws are often used to fell brush and small trees (<20.32 cm in diameter at breast height) and cut them into small pieces for hand-piling. Piles of woody biomass (called "slash" hereafter) are placed throughout the treated

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stand, with adequate spacing to reduce residual stand damage from pile burning. These piles are subsequently consumed by burning when weather conditions and the slash moisture content are within specified prescriptions.

Burning slash piles is a commonly used fuel treatment practice, but it is associated with several concerns including smoke production, residual tree mortality, and the risk of fire escape. These concerns limit the number of burning days, which often delays the completion of the fuel treatment and subsequent forest management activities. Furthermore it can cost $370–$2100/ha to implement, depending on the amount of fuel and the characteristics (species, size, and arrangement) of slash piles to be burned (Curran, 2008; personal communication, USDA Forest Service, California).

As an alternative to pile burning, mechanical removal of slash piles has been considered to avoid the negative effects and constraints associated with slash burning. At the same time it can also create opportunities to utilize slash to produce energy and bio-based products. Slash can be ground into hog fuel which is transported using wood chip vans to cogeneration/power plants to produce steam heat and/or electricity. Hog fuel is a term describing biomass fuel that has been processed by a mechanical shredder or grinder and is normally used as a fuel source for heat or energy production. It is a whole-tree product that includes bark, leaves, and stem wood.

Slash utilization can also include forest residues resulting from commercial thinning and timber harvesting operations. However, this approach has not been well implemented on a practical level because of limited accessibility on forest roads and the high costs associated with collection and transportation of these materials [5]. Typical slash recovery operations for energy production use a grinder and chip vans. These machines are brought to landings if chip vans can access these locations. Forest roads in the western US are built for hauling logs using stinger-steered logging trucks, and a typical chip van has limited access to harvesting sites due to sharp curves, steep grades, and low ground clearance. Because of limited accessibility, woody biomass amounts that can practically be supplied for utilization are substantially less than gross forest biomass available to harvest. The annual available biomass in California was estimated at 24.4 million dry metric tons, but the amount that could potentially be utilized amounts to only 13.0 million dry metric tons [8].

Besides limited accessibility for chip vans, high-cost machines such as grinders are significantly underutilized since landing slash piles are small (a grinder often spends less than one or two days grinding a pile of landing slash) and scattered over a large area. An expensive grinder (typically >$450,000) has to move frequently, resulting in low production rates. Logistical arrangements such as grinding location and hauling distance on forest roads between on-site operations and transportation to an energy plant have been a challenge as well. These difficulties are further complicated in northern California where terrain is steep and roads are typically not favorable for efficient transportation.

In this study we examined an alternative method to remove woody materials and increase accessibility to slash piled as a result of shaded fuelbreak treatments. In particular a ''roll-off truck'' with roll-on/off containers was tested to carry non-merchantable material for a short distance (less than 8 km) to a centralized processing site. ''Roll-off'' refers to a straight frame truck configuration in which modular containers are ''rolled'' onto and off of the straight frame truck by means of a truck-mounted hydraulic winch and a hook. A previous study indicated that a roll-off trucking system would significantly improve both accessibility to more forest residues and economic efficiency of the recovering process [7]. Further investigation is needed to broaden our knowledge for a wide range of applications of this technology into woody biomass collection and transportation.

The overall objective of this study was to evaluate the economic feasibility of removing hand-piled slash using a roll-off trucking system in mountainous conditions. Specific research questions include: 1) how much does it cost ($/hour) to operate the system consisting of roll-offs and a loader; 2) what are the potential productivities (tonnes/hour) in a wide range of work conditions for loading and transportation; 3) what is the economic maximum hauling distance in which a roll-off trucking system is financially feasible; and 4) what would it take to develop a profitable business that utilizes roll-off trucks in slash collection and transportation.

2. Study methods

2.1. *Study site and a roll-off trucking system*

A two-week trial of removing hand-piled slash using a roll-off trucking system was conducted on the Six Rivers National Forest in late July, 2007 near Mad River, California. Hand-piled slash (Fig. 1) was created by shaded fuelbreak treatments which also maintained 60% crown closure to reduce re-growth of brush and small trees within the understory. The shaded fuelbreak treatment prescription required cutting brush and suppressed understory trees less than 20.32 cm in diameter at breast height (DBH) which contribute to ladder fuels. These ladder fuels can allow fires to carry from surface fuels into the crowns of trees or shrubs with relative ease. The width of the shaded fuelbreak treatment varied with topography and vegetation but averaged 76.2 m on ridge tops and along roads.



Fig. 1 – Hand-piled slash from a shaded fuelbreak treatment. There were 198 slash piles per ha on average.

Firewood was salvaged from the trees cut and the remaining unmerchantable woody biomass was hand-piled for later burning. Realizing an opportunity for biomass research, however, a collaborative agreement was reached to collect and transport slash piles to a central processing area using a roll-off trucking system. Once at the processing area, slash could be ground and transported for use at an energy plant.

The slash removal system utilized in this study consisted of a loader and a roll-off truck equipped with roll-off containers (Fig. 2). Containers were placed along the roadside and loaded with slash in the woods using a small rubber tracked skid-steer front-end loader (ASV RC100). A typical loading cycle included unloaded travel to the slash pile, compiling/picking-up slash, loaded travel back to the container, and the subsequent loading of the container with the slash. This particular loader was chosen for use on this project due to its ability to travel swiftly around and without damaging residual trees while exerting minimal ground pressure (<27.58 kPa). The loader operator compacted the slash in the container using the bucket to increase the slash weight within the container. The loader is small enough to be transported inside a container when moving to the next job, but was carried to the work site using a large pickup truck ($40,000) and a trailer ($10,000).

A roll-off truck, utilizing its 317 kW engine, short truck length (9.1 m) and high clearance design, can negotiate sharp curves (~12.1 m radius) and travel on steep grades (15–20%) where typical log trucks can travel. A trailer used to carry an additional container (often referred to as a "pup") can be pulled by the truck as well. The roll-off containers used for this study were 2.1 m tall, 2.4 m wide, and 6.1 m length with a capacity of 30.6 m^3. Although the containers were similar, one had a non-removable stabilizer bar over the rear doors. With two containers, the truck can carry up to 18.2 tons of woody biomass per trip. The truck traveled on four different standards of roads during the study: paved highway, main gravel, secondary gravel, and spur (Table 1). Each trucking cycle consisted of travel to a loaded container, loading (rolling-on) a container filled with slash, loaded transport to a centralized processing area, dumping the slash from the container, transporting an empty container back to the loading area, and rolling off the empty container. Slash that was removed during the study was compiled at a central processing location where it was ground to hog fuel and delivered to a local energy plant; however, this grinding phase was not included in our study.



Fig. 2 – A roll-off trucking system consisting of a loader (above) and a roll-off truck equipped with roll-off containers (below).

2.2. *Data collection and analysis*

Machine rates ($/hour) for each machine were calculated using standard methods [3,6]. Cost factors and assumptions were collected from the contractors and summarized in Table 2. Time and motion methods were used to collect productive and non-productive time data for each machine. These data were used to examine interactions between equipment, personnel, and slash collection attributes (e.g. machine travel distance and number of slash piles). Equipment cycle times were recorded with one of two different stopwatches. Extremely short cycle elements observed during the loading cycles necessitated the use of a decimal stopwatch (hundredths of a minute), while longer trucking cycles allowed the use of a conventional stopwatch using minutes and seconds. Working conditions such as distances, weights, residual trees per acre, and pile size were measured and noted during each cycle element. Delays (i.e. machine not working) that occurred during the study were recorded and distributed within four categories: mechanical, operational, administrative, and personal delays. Research-related delays such as weight measurements were excluded from the production analysis.

A portable weighing system (Intercom PT 300) was used to measure slash weight carried in each container. The roll-off truck and two empty containers were weighed at the central processing site before the operation started to obtain a tare weight. After hauling the biomass to the central processing site, the truck and load were weighed. Weight measurements were recorded along each of the truck's three axles, and summed to obtain an overall weight. The slash weight was calculated by subtracting the empty truck weight including a container (tare weight) from the gross loaded weight.

Truck travel distances on paved and gravel roads, as well as at the central processing area were measured by vehicle odometer down to 0.16 km. Distances on spur roads within the treatment area were measured by fiberglass tape and/or impulse laser. Distances were marked and flagged along the spur road for convenient identification while collecting data. Locations where road types changed were flagged to identify them during data collection.

Table 1 – Characteristics of roads used to transport slash between slash loading sites and a central processing area.

	Paved highway	Main gravel	Secondary gravel	Spur[b]
One-way distance (km)	0.56	1.84	1.36	Up to 0.61 (or 660 m)
Road surface	asphalt paved	aggregate surfaced	aggregate[a] surfaced	native soils
# lanes	two	two	single	single
Road grade (%)	mean: 2	mean: 10	mean: 15	mean: 10
	min.: 0	min.: 2	min.: 2	min.: 2
	max.: 2	max.: 15	max.: 21	max.: 15
Curvature	low-boy accommodated	low-boy accommodated	low-boy not accommodated	low-boy not accommodated
Maximum design speed (kph[c])	72–88	24	16	8
Actual observed speed (kph[c])	56.3–64.4	16.1–19.6	12.8–16.1	3.2–9.6

a a low component of fine soils mixed with less than three inch size gravels.
b spur roads did include rough road surface, water bars, and windy curves.
c kilometer per hour.

Slash moisture measurements were recorded daily during the study in order to determine the average moisture content of the material being hauled. Data were collected for three diameter size classes: small (<2.5 cm), medium (6.7–15.2 cm), and large (>15.2 cm). Biomass was randomly selected from slash piles within the three size classes for measurement. Moisture samples were taken using a Delmhorst BD-2100 moisture meter. The moisture meter sensor penetrated through the bark on small-diameter material to measure moisture content. For medium and large material, pieces were cut using a chainsaw to obtain a cross-sectional area that was not exposed to ambient air. Three measurements were taken diagonally across each cross-sectional area (top, bottom, and center). Average moisture content was obtained by averaging the three measurements. Three to seven circular sample plots (0.04 ha in size) were established on 0.2–04 ha study units to determine the average number of piles and residual trees per hectare.

The spreadsheet program MS Excel was used to calculate simple statistics such as individual loading cycle times, and the number of loading cycles per loaded container. In addition, individual delay-free cycle times were calculated and paired with measured variables (e.g., distances, trees per acre, load size) observed during the cycle. Regression analysis was conducted through the SAS 9.0 program to determine if variables had a significant effect on loading and hauling productivity. Variables were considered significant if the p-value of the associated regression coefficient was less than 0.05: a variable which had a p-value greater than 0.05 was not included in the regression model. Before performing multiple regression analysis, the data were examined for outliers, normality, heteroscedasticity, and the severity of multicollinearity.

Sensitivity analysis was also performed to evaluate the maximum economic hauling distance on forest roads. This is useful to examine when the overall cost of slash removal becomes cost prohibitive for the Forest Service and/or private landowners. Hauling distance also directly affects the number of containers required to balance the workload between truck

Table 2 – Summary of cost factors and assumptions used to calculate hourly costs.

Cost factors	Roll-off truck	Loader	Pickup truck[a]
Purchase price ($)	98,000[b]	58,000	40,000
Salvage value (% purchase price)	20	20	10
Economic life (years)	7	5	7
Interest (% average yearly investment)	8.9	10	10
Insurance (% average yearly investment)	3	3	3
Taxes (% average yearly investment)	2	2	2
Repair & Maintenance (% depreciation)	65	75	50
Fuel consumption (liters/hour)	15	11	23 L/day[c]
Fuel cost ($/liter)	0.87	0.87	0.87
Lube and oil cost (% fuel cost)	37	37	37
Wages ($/hour)	20	23.40	0
Fringe benefits (% wage)	12	30	0
Scheduled machine hour (SMH)/year	1600	1600	1600
Productive machine hour (PMH)/year	1440	1360	1360
Utilization rate (%)	90	85	85
Machine cost ($/SMH)	55.82	56.38	9.43
Machine cost ($/PMH)	62.02	66.32	11.89[d]

a the pickup truck used to carry the loader to the study site and daily trip for workers.
b the price includes two 30.6-m^3 containers ($4000 each).
c fuel consumption for daily trip to the study site (36 km one-way distance).
d the machine cost includes the trailer used to carry the loader ($0.80/PMH).

and loader. For loading, our sensitivity analysis was focused on residual stand density and forest travel distance to understand how these variables affected loading productivity and cost.

A cash flow analysis that includes all the costs and revenues was conducted to evaluate the profitability of the contracting business over the life of the machines. The ChargeOut! [2] program was used to calculate internal rates of return, break-even charge-out rates, and net present values. This program also allows a user to perform sensitivity and break-even analysis to understand the business conditions that are necessary to make the business profitable, including the number of work days, loan interest, risk premium, and charge-out rates.

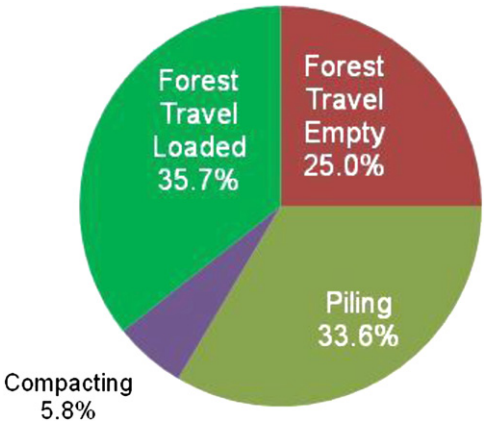


Fig. 3 – Elements of the loading cycle time. The average cycle time was 1.68 min.

3. Results and discussion

3.1. *Loading slash into roll-off containers*

The average cycle time for the loader to make a round trip between a container and the slash piles took 1.68 min (Table 3). The largest time component of an average cycle time (35.7%) was travel from the slash pile back to the container at an average traveling distance of 35.4 m (Fig. 3). Piling represented 33.6% of the average cycle time, indicating that a different type of grapple for the loader may be used to more efficiently grab the slash pile. Residual tree density often slowed loaded and empty travel time due to limited maneuverability and increased care needed to reduce residual stand damage. This was particularly noticeable when the number of trees per ha exceeded 494. In order to maximize the load volume, the operator combined two or three piles of slash using the grapple and bucket combination to compact and consolidate the piles. Approximately 1/3 of the overall delay-free cycle time was spent piling and compacting slash in the forest. Although additional slash piling and compacting activities increased the delay-free cycle time ($p < 0.05$), this process also increased the load size delivered to the container. Compaction of slash in the forest occurred during 85% of the total cycles observed ($n = 891$).

Since the loader had rubber tracks, there was a concern by the operator that the tracks could be seriously damaged by stumps left by the thinning crew. In addition to physical damage, traveling over stumps slowed travel time since the slash grappled was often unacceptably shaken loose. The loader operator removed the stumps during 7.7% of the overall cycles though it did not significantly contribute to individual cycle times. Although care was taken, by the end of the operation, the tracks had been severely damaged by stumps. The stump height should be reduced close to ground level if this type of loader is expected to be used to remove slash.

The average weight of a load varied with the amount of bole wood in the pile and averaged 202 kg ($n = 889$). The number of loading cycles needed to fill a 30.6-m^3 container averaged 17.5, ranging from 13 to 26. Small limbs and branches were difficult to compact in the container and caused lighter container loads. Areas with more bole material had more weight and compaction was achieved primarily by the weight of the material. Observations showed an increase in weight of 161% between the lightest container load (limbs and branches) and the heaviest container loaded with limbs, tops, and bole wood. Although eight additional loading cycles were needed to achieve this weight, the extra time spent was justified by the increased tonnage per container.

Regression models for delay-free cycle time were developed to estimate an average loading cycle time, using the corresponding independent variables for each cycle element (Table 4). For the models, all the coefficients were significant ($p < 0.05$) and R^2 values were generally high, indicating the average cycle time equation for loading may be effectively used to estimate loading productivity. Ground slopes were generally gentle (<32%) and did not significantly affect the average cycle time ($p > 0.05$).

Dummy variables were used to examine the effect of operators (i.e., experienced and less experienced), time for stump removal, and different container types. All dummy variables were significant, indicating that a less experienced

Table 3 – Summary statistics of observed loading cycle time and its variables ($n = 891$).

Loading cycle component	Variable	Mean	Standard deviation
Forest travel empty[a]	time (sec.)	0.41	0.19
	distance (m)	30.31	18.76
	TPH[e]	435	151.65
Piling[b]	time (sec)	0.56	0.40
	distance (m)	4.66	5.61
	# of piles	3.42	2.29
Forest travel loaded[c]	time (sec.)	0.59	0.28
	Distance (m)	34.52	19.52
	slope (%)	15.70	7.97
	TPH[e]	435	151.65
Compacting load[d]	time (sec.)	0.10	0.29
	cycle #[f]	10.13	5.85

a loader driving to the slash piles.
b loader piling slash to make a full load.
c loader traveling back to the container with a full load of slash.
d loader compacting to increase the slash weight in the container.
e trees per ha.
f the number of cycles in which the loader starts compacting the slash to increase the slash load weight.

Table 4 – Regression models for average loading cycle time (estimates of time in centi-minutes; $n = 891$).

Operation element	Regression model	R^2	Prob. > F	Std. Err.	*p*-value
Forest travel empty	time = 0.340	0.83	<0.0001	0.013	<0.0001
	+0.007 (distance in m)			0.003	<0.0001
	+0.073 (operator indicator[a])			0.011	<0.0001
Piling	time = 0.049	0.76	<0.0001	0.012	<0.0001
	+0.013 (distance in m)			0.013	<0.0001
	+0.124 (# of piles)			0.003	<0.0001
	+0.314 (remove stump indicator[b])			0.025	<0.0001
Forest travel loaded	time = 0.238	0.81	<0.0001	0.042	<0.0001
	+0.007 (distance in m)			0.003	<0.0001
	+0.008 (cycle # in loading a container)			0.002	<0.0001
	+0.0004 (TPH[d])			0.0004	<0.0001
	+0.109 (operator indicator[a])			0.019	<0.0001
Compacting load	time = 0.251	0.96	<0.0001	0.019	<0.0001
	+0.189 (cycle # in loading a container)			0.003	<0.0001
	+0.208 (operator indicator[a])			0.208	<0.0001
Combined	Total cycle time = −0.009	0.78	<0.0001	0.053	0.8620
	+0.006 (cycle # in loading a container)			0.002	0.0141
	+0.005 (slope in %)			0.002	0.0022
	+0.0004 (TPH[d])			0.0008	<0.0001
	+0.282 (operator indicator[a])			0.026	<0.0001
	+0.010 (forest travel empty distance in m)			0.020	<0.0001
	+0.016 (piling distance in m)			0.026	<0.0001
	+0.137 (# of pilings)			0.006	<0.0001
	+0.451(remove stump indicator[b])			0.045	<0.0001
	+0.258 (# of compacting)			0.010	<0.0001
	+0.090 (container indicator[c])			0.024	0.0002
	+0.007 (forest travel loaded distance in m)			0.016	0.0001

a indicator variable: 0 for experienced operator, or 1 for less experienced operator.
b indicator variable: 0 if stump is not removed, or 1 if stump is removed.
c indicator variable: 0 if container with bar is used, or 1 if container without bar is used.
d trees per ha.

operator and stump removal could increase delay-free average loading cycle time. This also indicates that container type has an effect on loading time: the container with a stabilizer bar could not be filled as high as the container without a stabilizer bar because the slash above the bar may cause hang-ups delaying the slash unloading process. Although a decrease in loading time was observed, there was an 18% decrease in tonnage when comparing the heaviest loaded containers.

3.2. *Hauling slash to a central processing area*

Once a container was loaded, the roll-off truck transported the slash to a central processing area located 4.5 km away from the loading site. Due to poor road conditions (i.e., steep and sharp curves), a round trip (8.9 km) took slightly over 52 min (Table 5). The average travel time to reach a loaded container along the spur road was approximately 2 min with an average travel distance of 91.4 m. At times, however, the truck had to back over 304.8 m to reach a loaded container due to highly limited turn-around locations on the spur road. Backing the truck long distances (particularly over steep or uneven terrain) could severely lower the production rate. The average total time to drive to a loaded container, position the truck, and load a full container was 6.79 min, representing 12.9% of the average total cycle time.

Delays were seldom observed during the loading stage. The truck driver was highly skilled at positioning the truck for optimum hooking and rolling the loaded container onto the truck. The major challenge associated with rolling-on and rolling-off containers was the uneven terrain located adjacent to the spur road. Uneven terrain did not allow the truck to have even vertical alignment on each rail, which at times, caused a container to drop off the rails (frames). When this occurred, extra time was needed to drop the container, and the likelihood of mechanical damage to the truck increased. Side-slopes should be avoided when loading/unloading containers. As the container is raised, the truck's center of gravity shifts to the side, increasing the chance of a rollover.

Travel speeds on spur roads were highly limited by steep grades and road surface conditions: the speeds on spur roads ranged from 3.2 to 9.7 km per hour (kph). Although the roll-off truck has almost 30-cm of ground clearance, extra time was spent traveling over road sections that contained rolling dips. The average time spent traveling over spur roads with a loaded container was 3.16 min (6.0 percent of the overall cycle time).

Traveling loaded on gravel roads accounted for 25.8 percent of the overall cycle time (13.49 min) and was constrained by sections with steep slopes (e.g. 21% slope), and poor road surface conditions (Table 6). Driving with a loaded container on downhill slopes required the assistance of an engine compression brake. Without the compression brake, applying the air brakes on loose, 7.6-cm or smaller rocks caused the truck to skid, lose steering ability, and caused an overall loss of vehicle control. Though effective, the compression brake contributed to reduced speed (travel loaded) on secondary gravel roads. Compared to the main

Table 5 – Summary of observed trucking cycle time and its independent variables ($n = 61$).

	Cycle elements	Distance (m)	Time		
			Minutes	%	Total %
Roll-on	Travel to container filled with slash	91	1.90	3.6	
	Positioning	49	1.36	2.6	
	Loading container filled with slash	N/A	3.53	6.7	12.9
Travel loaded	Spur road	307	3.16	6.0	
	Secondary gravel road	1368	7.16	13.7	
	Main gravel road	1851	6.33	12.1	
	Asphalt paved road	563	0.97	1.8	
	Central processing area	279	2.27	4.3	37.9
Unloading	Unloading slash	N/A	3.48	6.7	6.7
Travel empty	Central processing area	278	1.98	3.8	
	Asphalt paved road	563	0.99	1.9	
	Main gravel road	1851	6.93	13.2	
	Secondary gravel road	1368	6.40	12.2	
	Spur road	336	3.19	6.1	37.2
Roll-off	Drop empty container	N/A	2.79	5.3	5.3
Total		8905	52.44	100%	

gravel road, the average speed on the secondary gravel road was considerably slower due to loose surface conditions.

Field observations verified that containers with heavier payloads increased the travel speed on adverse gravel roads due to extra traction provided over the rear axles. This was particularly noticeable while traveling loaded on the main gravel roads. This, in combination with downhill traveling (loaded), increased the average speed on the main gravel road by 1.5 kph (Table 6). The load capacity (weight limit) for the roll-off truck was 9.1 tons, during the study, the slash weight per load ranged from 1.5 to 5.0 tons, with an average of 3.3 tons per load. This was far less than the load capability of the truck. Larger stem material loaded on top of lighter tops and limbs would compact the load and increase the weight (and volume) of each load.

Short travel distance on the paved road to the central processing site prevented the truck from reaching its normal speed. For 0.56 km of asphalt pavement, the loaded roll-off truck spent around 0.97 min with an average speed of 34.8 kph. A separate test of asphalt pavement travel over longer distance showed that the normal speed was around 64.4 kph. The travel speed of the roll-off truck at the central processing site (8.0 kph) was comparable with the speed on the gravel road due to short travel distance and limited road width along with a required stop for research related weight measurements. Traveling time on the paved road represented 4.3% of the total cycle time.

Unloading of slash at the central processing area involved positioning the truck at the proper location, opening the rear doors of the container, and raising the container to allow the biomass to slide out the back (similar to a dump truck). This operation was heavily affected by the design of the container. As noted previously, one of the containers had a bar on the back which hindered the smooth flow of slash out the back of the container. Occasionally slash would get caught on this bar preventing the slash from unloading. When this occurred, a chainsaw was needed to release the slash, significantly increasing the unloading time by 10–15 min. To mitigate the problems caused by the bar, the loader operator reduced the amount of slash loaded in this particular container. The average unloading time (delay-free) was 3.48 min, representing 6.7 percent of the average total cycle time.

A large central processing area is needed in order to accommodate large amounts of slash. Each slash load delivered comprised an area approximately 2.4 m wide by 6.1 m long after it was slid out of the roll-off container. By backing the empty container against the recently deposited pile, this footprint could be slightly reduced. The area should also

Table 6 – Average speed of the roll-off truck traveling on three different road sections. There was no significant difference ($p > 0.05$) of traveling speeds between travel empty and loaded on the paved road, but this was not the case for the gravel roads ($p < 0.05$).

Road sections	Distance (km)	Average travel speed (km/hour)	n
Pavement travel empty	0.56	34.3	61
Pavement travel loaded	0.56	34.9	68
Main gravel travel empty	1.84	16.1	61
Main gravel travel loaded	1.84	17.5	68
Secondary gravel travel empty	1.36	12.9	61
Secondary gravel travel loaded	1.36	11.4	68

Table 7 – Summary of regression analysis for trucking to carry roll-off containers.

Operation element	Regression model	n	R^2	Prob.>F	Std. Err.	p-value
Spur road loaded	time (sec) = 3.48	68	0.81	<0.0001	12.018	0.7732
	+158.70 (spur loaded direction[a])				13.171	<0.0001
	+0.52 (spur loaded distance in m)				0.033	<0.0001
Spur road empty	time (sec) = 13.91	60	0.80	<0.0001	14.056	0.3265
	+127.41 (spur empty direction[a])				10.005	<0.0001
	+0.39 (spur empty distance in m)				0.036	<0.0001
Site travel loaded	time (sec) = −139.68	69	0.49	<0.0001	44.935	0.0029
	+0.98 (site loaded distance in m)				0.161	<0.0001
Site travel empty	time (sec) = −305.81	60	0.72	<0.0001	34.675	<0.0001
	+1.54 (site empty distance in m)				0.125	<0.0001
Positioning	time (sec) = 85.23	60	0.98	<0.0001	6.129	<0.0001
	+0.62 (positioning distance in m)				0.016	<0.0001
Drive to loaded container	time (sec) = 100.97	60	0.91	<0.0001	12.177	<0.0001
	+0.39 (drive-to-box distance in m)				0.033	<0.0001
Combined	time (sec) = 355.88[b]	61	0.88	<0.0001	241.712	0.3413
	+0.52 (spur empty distance in m)				0.128	0.0001
	+131.76 (spur empty direction[a])				27.838	<0.0001
	+1.02 (positioning distance in m)				0.115	<0.0001
	+0.52 (travel to loaded container distance in m)				0.079	<0.0001
	+0.49 (spur loaded distance in m)				0.105	<0.0001
	+216.6 (spur loaded direction[a])				34.838	<0.0001
	+2.0 (site loaded distance in m)				0.813	0.0167

a Indicator variable: 0 for driving forward, or 1 for backing up the road.
b 355.88 = 588 − 232.12. −232.12 is the constant resulting from the combined regression analysis.

include sufficient room to maneuver the roll-off truck as well as a grinder, loader, and chip van.

Regression equations for delay-free trucking cycle elements were developed based on the data collected during the operation and contain the variables that significantly affected each elemental cycle time (Table 7). These regression models for each cycle element were combined to create a regression equation to estimate the average round-trip trucking time. This combined equation does not include the time required for loading/unloading containers, dumping the slash, and traveling all the road sections since cycle times spent on these operations were similar each cycle. The variable of "site travel empty" was not included in the combined regression model due to severe multicollinearity with the variable of "site travel loaded" and its insignificance ($p = 0.97$) in the model.

The combined regression equation did not include the average time of the following cycle elements; dropping off empty containers (167 s), loading a fully loaded container (212 s), and dumping biomass at the central processing site (209 s). To calculate an average cycle time, a constant value of 588 s was added to the combined regression equation to reflect these elements in the cycle. In addition, the time for traveling on gravel (main and secondary) and paved roads was not included in the combined equation because the truck traveled over the same (i.e., fixed) distances of those road sections for every trip. The average speed for each hauling road section (Table 6) was used to estimate travel time which should be added to the total cycle time.

3.3. *Collection and transportation cost*

The overall cost to collect and haul hand-piled slash which resulted from shaded fuelbreak treatments was $26.81/tons or $34.37bone dry metric ton (Table 8). Hourly cost ($/scheduled machine hour (SMH)) for the truck was lower than the hourly cost for the loader; however, the unit production cost ($/tonne) was higher in trucking than loading because hourly production rates for the truck were lower than those for the loader. This indicates that it is important to reduce trucking distance to minimize the overall cost. It should be noted that the average cycle time for trucking needs to be matched with the average cycle time for loading to avoid either truck or loader being idled. The difference in cycle times between loading and hauling may be also addressed by having multiple containers available.

Increasing hauling capacity for the truck is an option to keep the overall operation efficient. The truck is able to carry another container on a trailer, which would bring the combined total to 61.2 m^3 each trip. For example, under the

Table 8 – Productivity and cost of slash collection and transportation.

	Trucking	Loading
Machine cost ($/PMH[a])	62.02	78.22
Average cycle time (minutes)	52.15[b]	27.48[c]
Hourly productivity (tonne/PMH)	3.71	7.75
Cost ($/green ton[d])	16.71	10.10
Total cost	$26.81/green ton[d] or $34.37/bone dry metric ton	

a productive machine hour.
b time required to make a round trip including the time (9.80 min) for empty container dropping, picking up a fully loaded container and dumping slash at the central processing area.
c time required to fill a full load of slash in the 30.6-m^3 container.
d cost at 22% average moisture content.

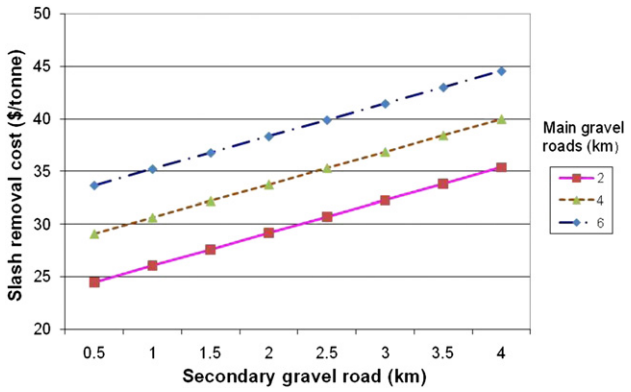


Fig. 4 – Effect of hauling distance on the total slash collection and transportation cost.

current cycle times for loading and hauling (Tables 3 and 5), it would be a somewhat balanced system if the truck were able to carry two containers each trip or another truck is added to the system since the loading time was about half of the trucking time. Two containers could be fully loaded with slash by the time the truck returns to the site from the central processing area. However, considerations should be given for extra time loading and unloading the second container and road conditions for truck turn-arounds if the truck and trailer option were implemented. Additionally, truck turnouts may be required if more than one truck were used.

A sensitivity analysis was performed to evaluate the effect of hauling distance on the total cost of slash collection and transportation; all values of variables were fixed while hauling distances for main and secondary gravel roads were changed on a 0.5-km increment. The regression equations developed from this study were used to estimate productivity for loading and trucking. The costs of roll-off trucking operations significantly increased with a small increase of hauling distance on gravel roads due to slow traveling speeds (less than 18 kph on gravel roads) and low carrying capacity (30.6 m^3 or 3.3 tons/trip in this study). In particular, the total slash removal cost increased at a faster rate when hauling distance on the secondary road increased, compared to the cost increase with an increase of hauling distance on the main gravel road (Fig. 4). This was mainly due to the truck's ability to travel faster on main gravel roads compared to secondary roads (Table 6). The cost increase for each 1 km increase of hauling distance on the main gravel roads was $2.30/tonne while it was $3.11/tonne for the secondary roads. Thus, it is important to understand the impact of hauling distance on the total slash removal cost.

3.4. *Financial analysis and business requirements for a roll-off trucking system*

The ChargeOut! [2] spreadsheet program was used to determine hourly break-even rates for each piece of equipment that would provide a required rate of return on invested capital (ROIC). ChargeOut! was also used to calculate internal rates of return (IRR). For our calculations we assumed a nominal (i.e., including inflation) deposit interest rate (4%) was added to an expected before tax risk premium (21%) to arrive at an expected nominal before tax required rate of return of 25%. Forest operations typically involve substantial business risk, requiring a relatively high risk premium, sometimes up to 25%. After adjusting for inflation (3%) and an income tax rate (33%), a real (i.e. inflation-adjusted) after tax return on invested capital (ROIC) of 13.4% was calculated. Depreciation was incorporated using the modified accelerated cost recovery method (MACRS) after allowing for a Section 179 deduction (Internal Revenue Service Publication 946, 2006). The Section 179 deduction allows an additional first-year write-off against taxable income in the first-year after the purchase of certain capital assets. The maximum Section 179 deduction was $108,000 in 2006. However, the firm must have sufficient income before depreciation that is at least equal to the total taxable write-off in order to take full benefit of the Section 179 deduction.

Based on our assumptions, the break-even charge-out rates for the truck and loader were $57.56/SMH and $69.54/SMH, respectively. At these rates, the contractors receive a 13.4% return on their invested capital after accounting for inflation and income taxes. The contractors would earn additional amounts beyond 13.4% return if they could negotiate charge rates higher than those break-even rates. These additional amounts may be used to cover cost items that are not included in the analysis such as overhead and other indirect cost items.

Positive cash flows over the life of the machine were shown after initial investments (Table 9), indicating that this business may be profitable as long as the input values correctly reflect the business conditions. Year 1 had the highest earning due to the assumption that the Section 179 deduction was

Table 9 – Summary of cash flows for trucking and loading in slash collection and transportation business. The annual operating hours was assumed to be 1600 SMH each year over the economic life of each machine.

	Year							
	0	1	2	3	4	5	6	7
Roll-off truck								
Cash flow before tax and financing	−99,000	20,766	20,169	20,774	21,397	22,039	22,700	43,181
Cash flow before tax	−59,400	10,925	10,328	10,933	11,556	12,198	22,700	43,181
Cash flow after tax	−59,400	37,818	4546	4732	4909	5076	15,209	28,931
Loader								
Cash flow before tax and financing	−59,500	16,527	15,193	15,648	16,118	28,501	–	–
Cash flow before tax	−35,700	7458	6124	6580	16,118	28,501	–	–
Cash flow after tax	−35,700	22,242	1492	1555	10,799	19,096	–	–

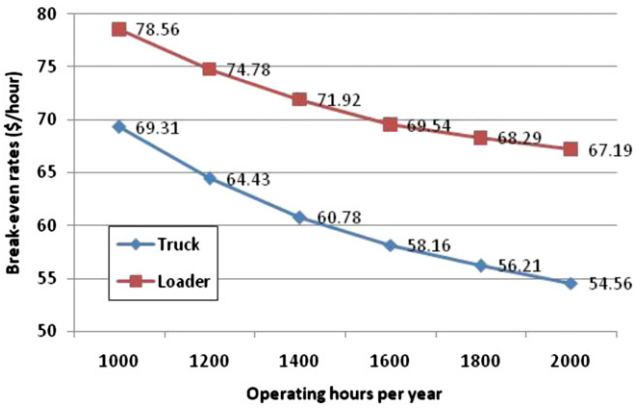


Fig. 5 – Effect of operating hours on break-even charge-out rates required to provide a 13.4% after tax real rate of return on investment. It was assumed that the repair and maintenance cost increases by 50% of the base rate after 4000 h of machine operation.

used against other taxable income. It was further noted that the last two years (3rd and 4th years for the loader; 6th and 7th years for the truck) also had high positive cash flows because they did not include any loan payments. It was assumed that loan payments were completed by the end of 3rd (loader) and 5th (truck) year.

One may wonder if there is enough work opportunity for the machines to generate revenue, which directly impacts the overall business profitability. A sensitivity analysis indicated that to earn at the specified ROIC, the break-even charge-out rates need to be higher with a decrease in operating hours per year (Fig. 5). For, example, the break-even rate should be increased to $79.03/hour for the loader and $70.02/hour for the truck if those machines were used only 1000 h per year. These represent 13.6% and 21.6% increases of the break-even rates, respectively, compared to the equivalent rates that were based on 1600 h per year.

The relationship is not perfectly linear because it is assumed that if the number of operating hours changes, the maintenance and repair costs will also change. In the ChargeOut! model, maintenance and repair costs do not have to be fixed. Rather, they may be allowed to change along with the number of annual productive hours or may be entered individually for each year.

4. Conclusion

A roll-off trucking system was effectively used to remove slash resulting from shaded fuelbreak treatments which was not accessible with a typical highway chip van. This system provides an option to access areas where current biomass harvesting systems, including highway chip vans, cannot be used because of poor access roads. This approach also allows for removal of hand-piled slash from inaccessible areas without pile burning, and helps create an opportunity for utilizing slash to generate energy or produce bio-based forest products. The overall cost to collect and haul hand-piled slash was $26.81/green ton with 22% average moisture content or $34.37/bone dry metric ton. The loader efficiently loaded slash into the containers at low cost ($10.10/tonne) while a roll-off truck hauled slash at $16.71/tonne for 4.5 km of forest road hauling distances. Mechanical removal of hand-piled slash would be an economically viable alternative to pile burning options under certain situations (e.g. short hauling distance and high slash volume per ha). Furthermore, slash burning options raise concerns related to air quality, risk of escaped fires, and limited burn opportunities.

It should be noted, however, that the costs of roll-off trucking operations could be significantly increased with increased hauling distance on gravel roads due to slow traveling speeds (less than 18 kph on gravel roads) and low slash weight being hauled (3.3 tons/trip on average). For example, the overall cost of slash removal using the roll-off trucking system increased by $3.11/tonne with each one-km increase in hauling distance over secondary gravel roads. This suggests that roll-off trucking systems should be used primarily for short hauling to a central processing area where sufficient access for highway chip vans is available.

The break-even charge-out rates for the truck and loader were $57.56/SMH and $69.54/SMH, respectively. At these rates, the contractors receive a 13.4% real rate of return on their invested capital after accounting for inflation and income taxes. These rates did not include overhead and other indirect cost items. Break-even charge rates could significantly increase with decreases in work days per year, but business investors should diversify work opportunities (e.g., waste management and construction sites) that could utilize a roll-off trucking system to keep their business profitable.

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